Part B:

Q2). Problems parameters:

States: 4×4 grid Coordinates (x, y) starting bottom-left at $(0,0)$

walls: $(0,3)$, $(1,1)$, $(3,1)$.

Sink States: Goal strictly at $(3,3)$ with $U = 1.0$.
Penalty at $(2,3)$ with $U = -1.0$.

Action Cost: -0.04 for all non-sink states.
$C(s,a)$

Discount factor (γ): 0.95.

Transition probability: Intended move $p = 0.7$,
Orthogonal moves $P_{perp} = \frac{1-0.7}{2} = 0.15$

Initilization $(t=0)$:

The sink states have fixed utilities: $U_0(3,3) = 1.0$

$U_0(2,3) = -1.0$

All other non-sink states are initialized to zero: $U_0(x,y) = 0$.

Iteration 1 $(t=1)$.

State $(0,0)$:

. Action UP: $0.7 \times U_0(0,1) + 0.15 \times U_0(0,0) + 0.15 \times U_0(1,0) = 0.7(0) + 0.15(0) + 0.15(0)$

$= 0$.

. Action DOWN: $0.7 \times U_0(0,0) + 0.15 \times U_0(0,0) + 0.15 \times U_0(1,0)$

$= 0.7(0) + 0.15(0) + 0.15(0)$

$= 0$.

. Action LEFT: $0.7 \times U_0(0,0) + 0.15 \times U_0(0,1) + 0.15 \times U_0(0,0)$

$= 0.7(0) + 0.15(0) + 0.15(0)$

$= 0$

Action Right $= 0.7 \times U_0(1,0) + 0.15 \times U_0(0,1) + 0.15 \times U_0(0,0)$.

$= 0.7(0) + 0.15(0) + 0.15(0)$

$= 0$.

Max EU: 0 (Action UP) $U_1(0,0) = -0.04 + 0.95 \times (0)$

$= -0.04$.

State (1,0):

Action UP $= 0.7 \times U_0(1,0) + 0.15 \times U_0(0,0) + 0.15 \times U_0(2,0)$
$= 0.7(0) + 0.15(0) + 0.15(0) = 0.$

Action DOWN $= 0.7 \times U_0(1,0) + 0.15 \times U_0(0,0) + 0.15 \times U_0(2,0)$
$= 0.7(0) + 0.15(0) + 0.15(0) = 0$

Action LEFT $= 0.7 \times U_0(0,0) + 0.15 \times U_0(1,0) + 0.15 \times U_0(1,0)$
$= 0.7(0) + 0.15(0) + 0.15(0) = 0$

Action Right $= 0.7 \times U_0(2,0) + 0.15 \times U_0(1,0) + 0.15 \times U_0(1,0)$
$= 0.7(0) + 0.15(0) + 0.15(0) = 0.$

Max EU : 0 (Action UP) $U_1(1,0) = -0.04 + 0.95(0) = -0.04.$

State (2,0).

Action UP $= 0.7 \times U_0(2,1) + 0.15 \times U_0(1,0) + 0.15 \times U_0(3,0)$
$= 0.7(0) + 0.15(0) + 0.15(0) = 0.$

Action DOWN $= 0.7 \times U_0(2,0) + 0.15 \times U_0(1,0) + 0.15 \times U_0(3,0)$
$= 0.7(0) + 0.15(0) + 0.15(0) = 0$

Action LEFT $= 0.7 \times U_0(1,0) + 0.15 \times U_0(2,1) + 0.15 \times U_0(2,0).$
$= 0.7(0) + 0.15(0) + 0.15(0) = 0$

Action Right $= 0.7 \times U_0(3,0) + 0.15 \times U_0(2,1) + 0.15 \times U_0(2,0).$
$= 0.7(0) + 0.15(0) + 0.15(0) = 0$

max EU : 0 (Action UP) $U_1(2,0) = -0.04 + 0.95 \times (0) = -0.04$

State (3,0):

Action UP $= 0.7 \times U_0(3,0) + 0.15 \times U_0(2,0) + 0.15 \times U_0(3,0)$

$= 0.7(0) + 0.15(0) + 0.15(0) = 0$

Action DOWN $= 0.7 \times U_0(3,0) + 0.15 \times U_0(2,0) + 0.15 \times U_0(3,0)$

$= 0.7(0) + 0.15(0) + 0.15(0) = 0$

Action LEFT $= 0.7 \times U_0(2,0) + 0.15 \times U_0(3,0) + 0.15 \times U_0(3,0)$

$= 0.7(0) + 0.15(0) + 0.15(0) = 0.$

Action DOWN $= 0.7 \times U_0(3,0) + 0.15 \times U_0(3,0) + 0.15 \times U_0(3,0).$

$= 0.7(0) + 0.15(0) + 0.15(0) = 0.$

Max EU: 0 (Action UP) $U_1(3,0) = -0.04 + 0.95 \times (0) = -0.04$

State (0,1)

Action UP $= 0.7 \times U_0(0,2) + 0.15 \times U_0(0,1) + 0.15 \times U_0(0,1)$

$= 0.7(0) + 0.15(0) + 0.15(0) = 0$

Action DOWN $= 0.7 \times U_0(0,0) + 0.15 \times U_0(0,1) + 0.15 \times U_0(0,1)$

$= 0.7(0) + 0.15(0) + 0.15(0) = 0$

Action LEFT $= 0.7 \times U_0(0,1) + 0.15 \times U_0(0,2) + 0.15 \times U_0(0,0)$

$= 0.7(0) + 0.15(0) + 0.15(0) = 0$

Action RIGHT $= 0.7 \times U_0(0,1) + 0.15 \times U_0(0,2) + 0.15 \times U_0(0,0).$

$= 0.7(0) + 0.15(0) + 0.15(0) = 0.$

Max EU: 0 (Action UP) $U_1(0,1) = -0.04 + 0.95 \times (0)$

$= -0.04.$

State (2,1):

Action UP : $0.7 \times U_0(2,2) + 0.15 \times U_0(2,1) + 0.15 \times U_0(2,1)$
$= 0.7 \times (0) + 0.15(0) + 0.15(0) = 0$

Action DOWN : $0.7 \times U_0(2,0) + 0.15 \times U_0(2,1) + 0.15 \times U_0(2,1)$
$= 0.7(0) + 0.15(0) + 0.15(0) = 0$

Action LEFT : $0.7 \times U_0(2,1) + 0.15 \times U_0(2,2) + 0.15 \times U_0(2,0)$
$= 0.7(0) + 0.15(0) + 0.15(0) = 0$

Action RIGHT : $0.7 \times U_0(2,1) + 0.15 \times U_0(2,2) + 0.15 \times U_0(2,0)$
$= 0.7(0) + 0.15(0) + 0.15(0) = 0$.

max EU : 0 (Action UP) : $U_1(2,1) = -0.04 \times 0.95 \times (0) = -0.04$

State (0,2):

Action UP : $0.7 \times U_0(0,2) + 0.15 \times U_0(0,2) + 0.15 \times U_0(1,2)$
$= 0.7(0) + 0.15(0) + 0.15(0) = 0$

Action DOWN : $0.7 \times U_0(0,1) + 0.15 \times U_0(0,2) + 0.15 \times U_0(1,2)$
$= (0.7 + 0.15 + 0.15)(0) = 0$

Action LEFT : $0.7 \times U_0(0,2) + 0.15 \times U_0(0,2) + 0.15 \times U_0(0,1)$
$= (0.7 + 0.15 + 0.15)(0) = 0$

Action Right : $0.7 \times U_0(1,2) + 0.15 \times U_0(0,2) + 0.15 \times U_0(0,1)$
$= (0.7 + 0.15 + 0.15)(0) = 0$.

Max EU : 0 (Action UP) : $U_1(0,2) = -0.04 + 0.95(0) = -0.04$

State (1,2)

Actions:

UP: $U_0(1,3)\,0.7 + U_0(0,2)\,0.15 + U_0(2,2)\,0.15$

$= (0.7+0.15+0.15)(0) = 0$

DOWN $= U_0(1,2)\,0.7 + U_0(0,2)\,0.15 + U_0(2,2)\,0.15$

$= (0.7+0.15+0.15)(0) = 0$

LEFT: $U_0(0,2)\,0.7 + U_0(1,3)\,0.15 + U_0(1,2)\,0.15$

$= (0.7+0.15+0.15)(0) = 0$

Right $= U_0(2,2)\,0.7 + U_0(1,3)\,0.15 + U_0(1,2)\,0.15$

$= (0.7+0.15+0.15)(0) = 0$

max EU: 0 (Action UP) $U_1(1,2) = -0.04 + 0.95(0) = -0.04$.

State (2,2)

Actions:

Up $= U_0(2,3)\,0.7 + U_0(1,2)\,0.15 + U_0(3,2)\,0.15$

$= (0.7+0.15+0.15)(0) = 0$.

DOWN $= U_0(2,1)\,0.7 + U_0(1,2)\,0.15 + U_0(3,2)\,0.15$

$= (0.7+0.15+0.15)(0) = 0$

LEFT $= U_0(1,2)\,0.7 + U_0(2,3)\,0.15 + U_0(2,1)\,0.15$

$= 0.7(0) + 0.15(-1) + 0.15(0) = -0.15$

DOWN $= U_0(3,2)\,0.7 + U_0(2,3)\,0.15 + U_0(2,1)\,0.15$

$= 0.7(0) + 0.15(-1) + 0.15(0) = -0.15$

Max EU: 0 (Action DOWN) $U_1(2,2) = -0.04 + 0.95 \times (0)$

$= -0.04$.

State (3,2):

Actions:

UP $= U_0(3,3)\,0.7 + U_0(2,2)\,0.15 + U_0(3,2)\,0.15 = 0.7(1) + 0.15(1)$

$= \cancel{(0.7+0.15+0.15)(0)} = 0.7$

DOWN $= U_0(3,2)\,0.7 + U_0(2,2)\,0.15 + U_0(3,2)\,0.15$

$= (0.7+0.15+0.15)(0) = 0$

LEFT $= U_0(2,2)\,0.7 + U_0(3,3)\,0.15 + U_0(3,2)\,0.15$

$= 0.7(0) + 0.15(1) + 0.15(0) = 0.15$

RIGHT $= U_0(3,2)\,0.7 + U_0(3,3)\,0.15 + U_0(3,2)\,0.15$

$= 0.7(0) + 0.15(1) + 0.15(0) = 0.15$

Max EU: 0.7 (Action UP) $U_1(3,2) = -0.04 + 0.95(0.7)$

$= 0.625$

State (1,3):

Actions:

UP $= U_0(1,3)\,0.7 + U_0(1,3)\,0.15 + U_0(2,3)\,0.15$

$= 0.7(0) + 0.15(0) + 0.15(-1) = -0.15$

DOWN $= U_0(1,2)\,0.7 + U_0(1,3)\,0.15 + U_0(2,3)\,0.15$

$= 0.7(0) + 0.15(0) + 0.15(-1) = -0.15$

LEFT $= U_0(1,3)\,0.7 + U_0(1,3)\,0.15 + U_0(1,2)\,0.15$

$= (0.7+0.15+0.15)(0) = 0$

RIGHT $= U_0(2,3)\,0.7 + U_0(1,3)\,0.15 + U_0(1,2)\,0.15$

$= 0.7(-1) + 0.15(0) + 0.15(0) = -0.7$

Max EU: 0 (Action LEFT) $U_1(1,3) = -0.04 + 0.95(0$

$= -0.04$

Iteration 2 (t=2) :

State (0,0) :

Actions:

Up = $U_1(0,1)\,0.7 + U_1(0,0)\,0.15 + U_1(1,0)\,0.15$
$= 0.7(-0.04) + 0.15(-0.04) + 0.15(-0.04) = -0.04$

Down = $U_1(0,0)\,0.7 + U_1(0,0)\,0.15 + U_1(1,0)\,0.15$
$= (0.7 + 0.15 + 0.15)(-0.04) = -0.04.$

LEFT = $U_1(0,0)\,0.7 + U_1(0,1)\,0.15 + U_1(0,0)\,0.15$
$= (0.7 + 0.15 + 0.15)(-0.04) = -0.04.$

RIGHT = $U_1(1,0)\,0.7 + U_1(0,1)\,0.15 + U_1(0,0)\,0.15$
$= (0.7 + 0.15 + 0.15)(-0.04) = -0.04.$

Max EU: -0.04 (Action UP) $U_2(0,0) = -0.04 + 0.95 \times (-0.04)$
$= -0.078$ //

State (1,0) :

Actions :

Up = $U_1(1,0)\,0.7 + U_1(0,0)\,0.15 + U_1(2,0)\,0.15$
$= (0.7 + 0.15 + 0.15)(-0.04) = -0.04.$

Down = $U_1(1,0)\,0.7 + U_1(0,0)\,0.15 + U_1(2,0)\,0.15$
$= (0.7 + 0.15 + 0.15)(-0.04) = -0.04$

LEFT = $U_1(0,0)\,0.7 + U_1(1,0)\,0.15 + U_1(1,0)\,0.15$
$= (0.7 + 0.15 + 0.15)(-0.04) = -0.04.$

Right = $U_1(2,0)\,0.7 + U_1(1,0)\,0.15 + U_1(1,0)\,0.15$
$= (0.7 + 0.15 + 0.15)(-0.04) = -0.04.$

Max EU: -0.04 (Action UP) $U_2(1,0) = -0.04 + 0.95(-0.04)$
$= -0.078$ //

State (2,0):

Actions:

UP $= U_1(2,1)0.7 + U_1(1,0)0.15 + U_1(3,0)0.15$

$= (0.7+0.15+0.15)(-0.04) = -0.04$

DOWN $= U_1(2,0)0.7 + U_1(1,0)0.15 + U_1(3,0)0.15$

$= (0.7+0.15+0.15)(-0.04) = -0.04$

LEFT $= U_1(1,0)0.7 + U_1(2,1)0.15 + U_1(2,0)0.15$

$= (0.7+0.15+0.15)(-0.04) = -0.04$

RIGHT $= U_1(3,0)0.7 + U_1(2,1)0.15 + U_1(2,0)0.15$

$= (0.7+0.15+0.15)(-0.04) = -0.04$

Max EU $= -0.04$ (Action UP) $U_2(2,0) = -0.04 + 0.95 \times (-0.04)$

$= -0.078$

State (3,0):

Actions:

UP $= U_1(3,0)0.7 + U_1(2,0)0.15 + U_1(3,0)0.15$

$= (0.7+0.15+0.15)(-0.04) = -0.04$

DOWN $= U_1(3,0)0.7 + U_1(2,0)0.15 + U_1(3,0)0.15$

$= (0.7+0.15+0.15)(-0.04) = -0.04$

LEFT $= U_1(2,0)0.7 + U_1(3,0)0.15 + U_1(3,0)0.15$

$= (0.7+0.15+0.15)(-0.04) = -0.04$

RIGHT $= U_1(3,0)0.7 + U_1(3,0)0.15 + U_1(3,0)0.15$

$= (0.7+0.15+0.15)(-0.04) = -0.04$

Max EU $= -0.04$ (UP) $U_2(3,0) = -0.04 + 0.95(-0.04)$

$= -0.078$

State (0,1): Actions:

UP $= U_1(0,2)\,0.7 + U_1(0,1)\,0.15 + U_1(0,1)\,0.15$
$= (0.7+0.15+0.15)(-0.04) = -0.04$

DOWN $= U_1(0,0)\,0.7 + U_1(0,1)\,0.15 + U_1(0,1)\,0.15$
$= (0.7+0.15+0.15)(-0.04) = -0.04$

LEFT $= U_1(0,1)\,0.7 + U_1(0,2)\,0.15 + U_1(0,0)\,0.15$
$= (0.7+0.15+0.15)(-0.04) = -0.04$

Right $= U_1(0,1)\,0.7 + U_1(0,2)\,0.15 + U_1(0,0)\,0.15$
$= (0.7+0.15+0.15)(-0.04) = -0.04.$

Max EU: -0.04 (UP) $U_2(0,1) = -0.04 + 0.95(-0.04)$
$= -0.078.$

State (2,1): Actions:

UP $= U_1(2,2)\,0.7 + U_1(2,1)\,0.15 + U_1(2,1)\,0.15$
$= (0.7+0.15+0.15)(-0.04) = -0.04$

DOWN $= U_1(2,0)\,0.7 + U_1(2,1)\,0.15 + U_1(2,1)\,0.15$
$= (0.7+0.15+0.15)(-0.04) = -0.04$

LEFT $= U_1(2,1)\,0.7 + U_1(2,2)\,0.15 + U_1(2,0)\,0.15$
$= (0.7+0.15+0.15)(-0.04) = -0.04$

Right $= U_1(2,1)\,0.7 + U_1(2,2)\,0.15 + U_1(2,0)\,0.15$
$= (0.7+0.15+0.15)(-0.04) = -0.04.$

Max EU: -0.04 (UP).

$U_2(2,1) = -0.04 + 0.95(-0.04)$
$= -0.078.$

State (0,2) Actions:

UP $= U_1(0,2)\,0.7 + U_1(0,2)\,0.15 + U_1(1,2)\,0.15$

$= (0.7 + 0.15 + 0.15)(-0.04) = -0.04$

DOWN $= U_1(0,1)\,0.7 + U_1(0,2)\,0.15 + U_1(1,2)\,0.15$

$= (0.7 + 0.15 + 0.15)(-0.04) = -0.04$

LEFT $= U_1(0,2)\,0.7 + U_1(0,2)\,0.15 + U_1(0,1)\,0.15$

$= (0.7 + 0.15 + 0.15)(-0.04) = -0.04$

RIGHT $= U_1(1,2)\,0.7 + U_1(0,2)\,0.15 + U_1(0,1)\,0.15$

$= (0.7 + 0.15 + 0.15)(-0.04) = -0.04.$

max EU $= -0.04$ (UP) $U_2(0,2) = -0.04 + 0.95 \times (-0.04) = -0.07$

State (1,2): Actions

UP $= U_1(1,3)\,0.7 + U_1(0,2)\,0.15 + U_1(2,2)\,0.15$

$= (0.7 + 0.15 + 0.15)(-0.04) = -0.04$

DOWN $= U_1(1,2)\,0.7 + U_1(0,2)\,0.15 + U_1(2,2)\,0.15$

$= (0.7 + 0.15 + 0.15)(-0.04) = -0.04$

LEFT $= U_1(0,2)\,0.7 + U_1(1,3)\,0.15 + U_1(1,2)\,0.15$

$= (0.7 + 0.15 + 0.15)(-0.04) = -0.04.$

RIGHT $= U_1(2,2)\,0.7 + U_1(1,3)\,0.15 + U_1(1,2)\,0.15$

$= (0.7 + 0.15 + 0.15)(-0.04) = -0.04$

max EU: -0.04 (UP) : $U_2(1,2) = -0.04 + 0.95(-0.04)$

$= -0.078.$

State (2,2) : Actions:

UP $= U_1(2,3)0.7 + U_1(1,2)0.15 + U_1(3,2)0.15$

$= 0.7(1) + 0.15(-0.04) + 0.15(0.625) = -0.61225$

DOWN $= U_1(2,1)0.7 + U_1(1,2)0.15 + U_1(3,2)0.15$

$= 0.7(-0.04) + 0.15(-0.04) + 0.15(0.625) = 0.05975$

LEFT $= U_1(1,2)0.7 + U_1(2,3)0.15 + U_1(2,1)0.15$

$= 0.7(-0.04) + 0.15(-1) + 0.15(-0.04) = -0.184$

Right $= U_1(3,2)0.7 + U_1(2,3)0.15 + U_1(2,1)0.15$

$= 0.7(0.625) + 0.15(-1) + 0.15(-0.04) = 0.2815$

max EU $= 0.2815$ (RIGHT) $U_2(2,2) = -0.04 + 0.95(0.2815)$

$= 0.22742$

State (3,2): Actions

UP: $= U_1(3,3)0.7 + U_1(2,2)0.15 + U_1(3,2)0.15$

$= 0.7(1) + 0.15(-0.04) + 0.15(0.625) = 0.78775$

DOWN $= U_1(3,2)0.7 + U_1(2,2)0.15 + U_1(3,2)0.15$

$= 0.7(0.625) + 0.15(-0.04) + 0.15(0.625) = 0.52525$

LEFT $= U_1(2,2)0.7 + U_1(3,3)0.15 + U_1(3,2)0.15$

$= 0.7(-0.04) + 0.15(1) + 0.15(0.625) = 0.21575$

RIGHT $= 0.7 \times U_1(3,2) + U_1(3,3)0.15 + U_1(3,2)0.15$

$= 0.7(0.625) + 0.15(1) + 0.15(0.625) = 0.68125$

MAX EU: 0.78775 (UP)

$U_2(3,2) = -0.04 + 0.95(0.78775)$

$= 0.70836$

State (1,3): Actions:

UP $= U(1,3)\,0.7 + U_1(1,3)\,0.15 + U_1(2,3)\,0.15$

$= 0.7(-0.04) + 0.15(-0.04) + 0.5(-1) = -0.184$

DOWN $= U_1(1,2)\,0.7 + U_1(1,3)\cdot 0.15 + U_1(2,3)\,0.15$

$= 0.7(-0.04) + 0.15(-0.04) + 0.15(-1) = -0.184$

LEFT $= 0.7 \times U_1(1,3) + 0.15 \times U_1(1,3) + 0.15 \times U_1(1,2)$

$= 0.7(-0.04) + 0.15(-0.04) + 0.15(-0.04)$

$= -0.04$

RIGHT $= U_1(2,3)\,0.7 + U_1(1,3)\,0.15 + U_1(1,2)\,0.15$

$= 0.7(-1) + 0.15(-0.04) + 0.15(-0.04)$

$= -0.712.$

Max EU: -0.04 (LEFT).

$U_2(1,3) = -0.04 + 0.95(-0.04)$

$= -0.078$ //.

So, These are the manually calculated ~~value~~ Utility values for $t = 1$ and $t = 2$ and perfectly matches the output (from python code).